ENSURING QUALITY PRODUCT
By Quality Function Deployment

This article describes how to get the framework on product requirements, basic design parameters, a conceptual clarity of various manufacturing processes required, and the quality assurance process. The final product design will, however, require further detailing. It will also require analysis of cost, technology, and business viability

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In my article titled 'Developing Lovable Products to Beat Competition' published in January issue I discussed lovable products. A product needs to have certain qualities for being loved. But quality is a very elusive concept. We all understand quality as something that makes the object desirable.

Beyond this universally accepted concept, quality means different things to different people. Some customers may give more priority to the looks, some may prefer more reliability, some others may look for a lightweight product, whereas another group may want one that costs less. To cater to these different needs the product has to address multiple things to be successful in the market.

Sometimes, we may need different product models to address these needs. In this article, we shall focus on the process of systematically addressing these varied requirements from a product.

There are various approaches for systematic product requirement analysis. We shall discuss a rather simple and popular technique called quality function deployment, or QFD. This is a process of listing the requirements simply and concisely, comparing various benchmark products, and systematically evolving the product definition and its manufacturing process.

For QFD we start with listing the product requirements. These requirements for a product come from various stakeholders. Part of it is from the people who will buy the product, then we have those who will use the product. Apart from these two groups we also need to address the requirements of those who sell, stock, and transport the product, the need of society at large, and the rules of the land.

If we fail to address all these varied needs, we shall end up with a bad quality product. So, our journey for product development has to start with discovering and documenting all these requirements.

The QFD process uses a set of four charts to systematically evolve the product definition, manufacturing, and quality assurance plans. The first chart is used to evolve product specifications to address the product needs. In this chart (Fig. 1) we list the product requirements in the first column.

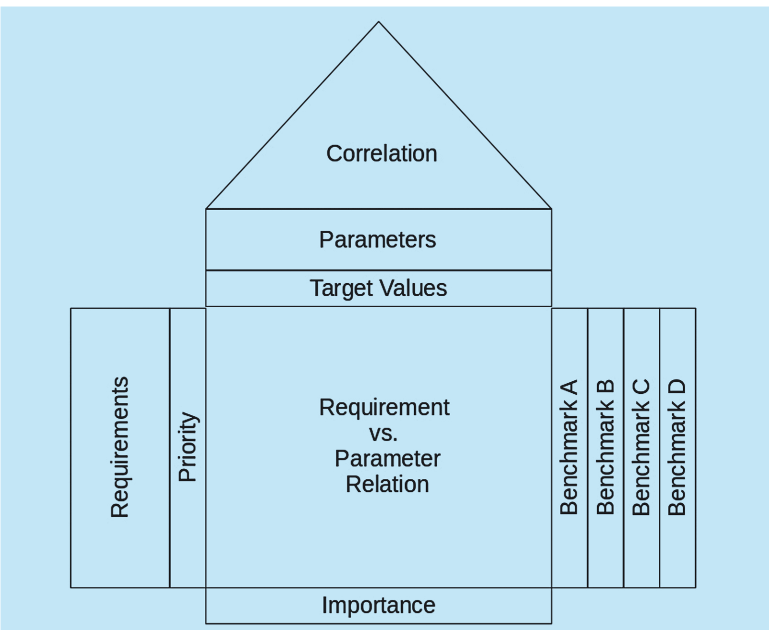

Fig. 1: QFD chart structure

Now, all requirements are not equally